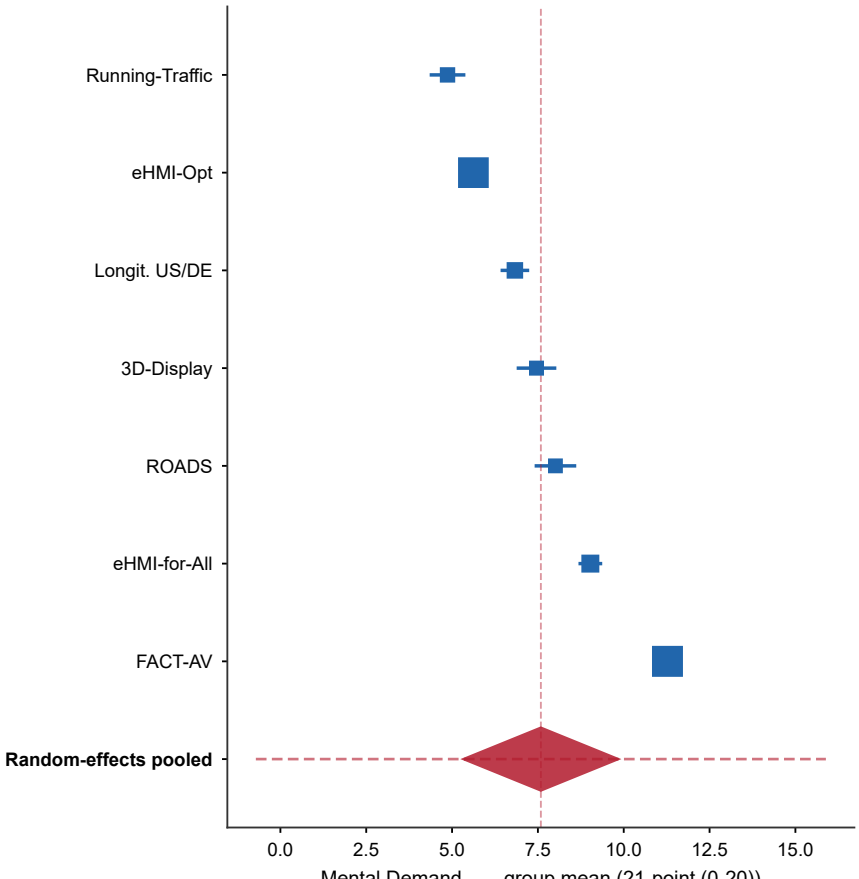
Evidence synthesis unlocked by standardization: NASA-TLX Mental Demand

Pooling is only possible because heterogeneous source labels were mapped to one canonical DV. Very high I² shows that shared names still mask scale/population differences.



	N obs	Mean	Wt	95% CI
Running-Traffic	288	4.87	4.3%	[4.3, 5.4]
eHMI-Opt	1,480	5.62	36.2%	[5.4, 5.8]
Longit. US/DE	470	6.83	6.6%	[6.4, 7.2]
3D-Display	45	7.46	3.5%	[6.9, 8.0]
ROADS	276	8.01	3.1%	[7.4, 8.6]
eHMI-for-All	720	9.03	9.7%	[8.7, 9.4]
FACT-AV	2,600	11.27	36.7%	[11.1, 11.5]
Random-effects pooled	**5,879**	**7.59**	**100%**	**[5.3, 9.8]**

Random-effects (DerSimonian–Laird), k = 7 studies. Heterogeneity: I^2 = 99.7%, τ^2 = 9.12, Q(6) = 2183, p < 0.001
Hartung–Knapp–Sidik–Jonkman 95% CI (t, 6 df): [5.6, 9.6]